

**DEPARTMENT OF TRANSPORTATION
OFFICE OF THE ASSISTANT SECRETARY FOR RESEARCH AND TECHNOLOGY
VOLPE NATIONAL TRANSPORTATION SYSTEMS CENTER (VOLPE CENTER)
ENVIRONMENTAL MEASUREMENT AND MODELING DIVISION, V324**

SOLE SOURCE JUSTIFICATION

A. Description of Supplies/Services. The Department of Defense (DoD) Air Force Civil Engineer Center (AFCEC) is the tactical center of excellence in regard to operational noise measurement and modeling for the United States Air Force. NOISEMAP is a suite of noise models used by the DoD to model aircraft noise that includes the Advanced Acoustic Model (AAM), Noise Map (NMAP), the Military Operating Area and Range Noise Model (MRNMAP), the Air Gunnery Model (AGM), the Rotorcraft Noise Model (RNM). These noise models are accessed via graphical user interfaced called BaseOps. BaseOps is a critical component for the accurate and efficient use of the NOISEMAP suite and AFCEC needs improvements to BaseOps import/export functionality and to its input checks in order to further improve efficiency and accuracy.

The Volpe Center is the primary department of transportation agency that develops transportation related noise models needed to ensure the unobstructed utilization of the United States' transportation infrastructure. In this role, the Volpe Center provides technical support to other agencies, including the Federal Aviation Administration, the Federal Highway Administration and the Department of Defense (DoD). In particular, the Volpe Center is supporting the DoD AFCEC in modeling military transportation related noise in order to meet AICUZ, NEPA and other mandates and statutes in order to remove obstacles to general flight operations and flight training (needed for military readiness) around installations. This current task aims to add new functionality to BaseOps to efficiently import and export data for military training exercises and to reduce conflicts between exclusion zones and air operations. BaseOps is a fundamental part of the DoD workflow for these analyses. Because this software is owned by Wasmer Consulting, changes to the software must be made by them.

The Contractor shall provide and updated version of BaseOps with appropriate documentation that will add Import/Export functionality for Airspace Data in a manner similar to the existing functionality for Flight and Static Data. It will also include an update to add a user warning when using the Check Case functionality that would warn the user when an Avoidance Area/MTR intersection existed. Details for each of these updates are given in the Software Requirements Specification. Given the level of detail described, the contractor will discuss with the Volpe Center any questions related to the specifications. The contractor will provide a beta version for testing by the Airforce, and, upon approval, a final version uploaded to the DENIX website along with updated documentation.

B. Market Research. The BaseOps user interface required by the NOISEMAP software is proprietary to Wasmer Consulting.

The data inputs for each of the noise models in NOISEMAP include an extremely complicated collection of aircraft types, engine settings, flight paths, etc., and the formats for these data demand a level of precision that, if done manually, would require an order or more magnitude of time to complete and would likely introduce otherwise avoidable human error. Automating much of the collection, processing and formatting of these input data greatly improves modeling efficiency and accuracy. To date, export functionality exists for fixed flight profiles but not for areas of operation. This update will improve the efficiency and accuracy of this common modeling task.

Research was performed by the technical lead at the Volpe Center on this work, Dr. Aaron Hastings. Dr. Hastings reviewed past contracting actions involving BaseOps at the Volpe Center and found that BaseOps is proprietary to Wasmer Consulting. Any work done with BaseOps source code or functionality testing can only be performed by Wasmer Consulting, and access to Wasmer Consulting is only possible through direct contract at this time.

C. Supporting Rationale.

(1) Sole Source: Only one responsible source and no other supplies or services will satisfy agency requirements.

- (i) This BaseOps update builds upon detailed work completed as part of the original BaseOps development. Wasmer Consulting was the sole organization performing the original development and the tool is proprietary to them.

- (ii) Not contracting with Wasmer Consulting to perform the required BaseOps update would deprive the Volpe Center, DoD AFCEC, and the military aviation community of the opportunity to improve a necessary tool and generate a more efficient method of optimizing military aircraft training opportunities. The Volpe Center risks the following consequences if Wasmer Consulting is not contracted to perform the BaseOps update described in the SOW:
- Longer modeling times for each noise study required for statutory requirements, installation maintenance, community outreach and operations planning.
 - Increased risk of modeling errors and longer QA/QC times due to the complexity of the model inputs.
- (iii) An optimistic time estimate for another contractor, working with both AFCEC and the Volpe Center, to develop a new user interface that includes the new desired features is three years. The estimated cost impact of the new contractor creating the new user interface, including the required updates, is \$1,000,000.

(2) Brand Name: N/A.

(3) Unusual and Compelling Urgency: N/A.

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